

Program Designing 'and' { [Programming].ICT 1305

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{ Getting to know you }

Chapter Alignment {

01 Course Lineup

02 Learning Outcomes

03 Evaluation Criteria

04 Sharing Materials

05 References

06 Introduction

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Course Lineup {



Techniques of Problem Solving

< Algorithms, Flowcharts, Pseudo Codes, Paper Simulations.>



Fundamentals of Programming

< Structure of a Program, Input/Output, Variable Declaration, Arithmetic Operations, Relational Operations, Logical Operations.>



Control Structures

< Conditional statements, Repetition statements , Break and Continue statements, Sub programs.>



Data handling

< Scope of Variables and Parameters, Recursion, Arrays, Records, Comparison of structure and class, Dynamic memory allocation, Best programming practices.>

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Learning Outcomes {

'At the end of the course, the you should be able to,'

<p> Demonstrate the knowledge of programming fundamentals </p>

<p> Demonstrate a basic knowledge in programming best practices </p>

<p> Design, write and test programs according to a basic requirement specification </p>

}

Evaluation Criteria {

End Exam  60%

< Written Exam > 30%

< Practical Exam > 30%

Assessments  40%

< Assignments >

< Quizzes >

< Practical tasks >

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“ Any Questions ??? ”

{ Contact :

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References {



< REF Problem Solving and Program Design in C 8th Edition - Jeri Hanly - Pearson />



< REF Simple Program Design - A Step by Step Approach - 4th Edition - L. Anne Roberson />



< REF How to Design Programs - 2nd Edition - Robert Bruce, Matthew Flatt />

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{

A farmer has some chickens and some goats.
Together there are 43 heads and 108 legs. How
many chickens does the farmer have? How many
goats?

}

What is a Problem {

< A problem = a situation with a desired outcome and a current state />

Examples:

- Sort a list of numbers
- Find a missing number in an array
- Navigate a maze

}

Problem Solving {

< Problem solving, or breaking down the solution to a problem into sequential steps is by far the most difficult part of computer programming.

There are many activities we can engage in that will help sharpen our problem solving skills. />

}

Why Problem Solving Matters {

- Programming is about solving problems systematically. Good problem-solving skills → better algorithms, debugging, code efficiency.
- Helps You Break Big Tasks Into Smaller Steps.
- Helps You Design Good Algorithms
- Leads to Cleaner, More Efficient Code

}

Why Problem Solving Matters {

- Helps You Debug Errors
- Improves Creativity in Solutions
- Essential for Working in Teams
- Makes You a Better Programmer Overall

}

Steps of Problem Solving {

- Identify the Problem
- Understand the Problem
- Define the Problem Clearly
- Generate Possible Solutions
- Evaluate and Compare Solutions
- Choose the Best Solution
- Implement the Solution
- Test and Review the Results

}

Strategies for Problem Solving {

- Decomposition – Break problem into smaller parts
- Work Backwards
- Pattern Recognition
- Use Trial and Error
- Make a Model or Diagram
- Use Logical Reasoning
- Brainstorm Multiple Solutions
- Create a Step-by-Step Plan (Algorithm)

}

Problem Solving & Programming {

“ How Problem Solving Connects to Programming ”

1. Problems inspire programs
2. Problem solving comes before coding
3. Programming uses logic—not guesswork
4. Algorithms are the bridge between thinking and coding
5. Debugging is problem solving
6. Efficient problem solving leads to efficient programs

}

“ Programming isn't about
typing, it's about thinking. ”

< Rich Hickey />

The Waiter {



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The Waiter {

- Three men in a cafe order a meal the total cost of which is \$15. They each contribute \$5.
- The waiter takes the money to the chef who recognizes the three as friends and asks the waiter to return \$5 to the men.
- The waiter is not only poor at mathematics but dishonest and instead of going to the trouble of splitting the \$5 between the three he simply gives them \$1 each and pockets the remaining \$2 for himself.
- Now, each of the men effectively paid \$4, the total paid is therefore \$12. Add the \$2 in the waiters pocket and this comes to \$14.....where has the other \$1 gone from the original \$15?

Three Bulbs {



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Three Bulbs {

- You are standing next to three switches. You know these switches belong to three bulbs in a room behind a closed door – the door is tight closed, and heavy which means that it's absolutely impossible to see if any bulb is on or not.
- All three switches are now in position off.
- You can do whatever you want with the switches and when you are finished you open the door and go into the room.
- While in there you have to tell which switch belongs to which bulb.
- How will you do that?

The Poisoned Wine {



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The Poisoned Wine {

- A king has 1000 wine bottles. One is poisoned.
- The poison kills within 10 hours.
- King has 10 prisoners and only one day to determine the poisoned bottle.
- How can you identify the bottle using the prisoners?

}

The Timer {

- You have two ropes that burn in exactly 60 minutes each – but burn unevenly.
- How do you measure 45 minutes?

}



The Weighing Jug {

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< You have a 3-liter and a  
5-liter jug. You need  
exactly 4 liters. />
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}
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Miscellaneous Problems {

< Two trains 100 miles apart travel toward each other at 20 mph and 30 mph. A bird starts at the first train, flying toward the second at 50 mph. It flies back and forth between the trains until they meet. How far does the bird travel? />

}

Miscellaneous Problems {

< A mechanic's brother won the car race. But the man who won had no brother. How is this possible? />

}

Miscellaneous Problems {

```
< Two fathers and two sons go fishing. Each catches  
one fish, but the total caught is only three. How  
is it possible? />
```

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}
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{ Thank You }